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A Deep Learning-based Transfer Learning Approach Fine-Tuned for Detecting Diabetic Retinopathy

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Abstract

Diabetics have blood sugar levels that are abnormal. Because of this, their eyesight can be affected. Untreated diabetes has the potential to result in permanent blindness, a grave complication. Among the severe consequences, early detection stands out as a critical hurdle for ensuring effective treatment. Regrettably, accurately assessing fundus images relies on human expertise to determine the stage of diabetic retinopathy. A simplified detection process could greatly benefit a vast population. Leveraging convolutional neural networks (CNN), which have demonstrated efficacy across various domains, diabetic retinopathy identification has seen advancements. The possibility of preventing complete, irreversible blindness lies in timely identification, underscoring the need for a dependable screening mechanism. To enhance the efficiency of screening and reduce errors that plague current models, a novel system has been introduced, capable of categorizing distinct diabetic retinopathy stages. This study focuses on examining datasets of Indian diabetic retinopathy images, encompassing four classes: Microaneurysms (MA), Soft Exudates (SE), Hard Exudates (EX), and Hemorrhages (HE). By employing the proposed VGG16 architecture for feature extraction and the Logistic Regression classifier for classification, an impressive accuracy of 90.4% was achieved.

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Keywords: Deep Convolutional Networks; Confusion Matrix; Machine Learning (ML) Algorithms; Transfer Learning; Diabetic Retinopathy (DR); Visual Geometry Group16.

1. Introduction

Diabetic Retinopathy is a disease that can affect a person's eye with diabetes who has had it for a long time and destroy the retina (DR). In the last 20 years, this disease, which is very frequent among diabetics, has

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affected 80% of those who have the condition. There is a 75% risk that they will go blind between the ages of 20 and 64. 347 million people worldwide suffer from DR, according to the World Health Organization (WHO). Worldwide, 366 million adults have diabetes, according to the International Diabetes Federation. By 2030, this number is anticipated to reach 552 million. According to a 2000 survey, diabetic retinopathy is very common in India (31.7 million cases), China (20.8), and the USA (17.7 million) [1]. Ophthalmologists' standard photographic technique is a colour fundus image. It takes a lot of effort and requires educated clinicians to manually evaluate colour fundus images of the retina and detect DR. Due to a shortage of skilled physicians and an understanding of how to use sophisticated equipment in remote areas, this becomes exceedingly difficult [2]. The expanding population of diabetics can be addressed by improving the infrastructure. Detailed planning and domain knowledge are required for traditional DR image classification systems. The key objective is to annotate the images, which calls for the expertise of experienced ophthalmologists. There are many frequently used machine learning (ML) techniques to forecast or classify a variety of health-related diseases [3] [4] [5]. Neural networks have been successfully used in the analysis and decision-making of drug discovery, medical image processing, and computer vision. By assisting in pathological screening and disease prediction, the application of these cutting-edge machine-learning approaches has significantly reduced the burden of human interpretations. Given their outstanding successes in various other healthcare fields, the employment of CNN and ML in the detection of diabetic retinopathy was a natural and inescapable focus of interest to decrease the incidence of diabetic retinopathy. The yearly ImageNet and COCO contests are just two of the several cutting-edge image classification projects that utilize CNNs nowadays [6][7].

2. Literature survey

Attia et al. [8] study on DR classification methods gave both conventional methods and deep learning strategies similar weight. Gupta and Chhikara [9] used various machine learning-based traditional techniques to evaluate the performance of DR detection and also exposed their shortcomings. These comparisons are based on the fundus image quality due to the poor contrast and image quality of some publically accessible datasets. Valarmathi and Vijayabhanu [10] investigated various recently used state-of-the-art CNN modifications for DR classification and drew attention to the inconsistent assessment measures applied to models in the literature. In their article [11], Using transformers for medical imaging functions such as segmentation, classification, and detection and reconstruction is described by Shamshad et al. The analysis found that, with more than 40 recent papers, research on transformer-based medical imaging reached a high around December 2021. The study also shows that between the duration of 2012 to 2015 approximately 27% of published papers used CNNs for segmentation tasks, and publications published in 2021 used vision transformers in 73% of cases. This implies a growing requirement for transformer-based segmentation methods. [12] analyses a few Vision Transformer(ViT) papers that focus on DR grading and classification with lesion identification concerning retinal diseases. In [13], authors used CNN to classify the various stages of DR. Images from the 500 retinal images in the Kaggle data set were used. The HE filtering method is used along with the stochastic gradient descent with a momentum optimization approach before training. The proposed model has an accuracy of 81.1% and combines the VGG 16 Alex Net with the Inception Net V3 to determine the result. Transfer learning provides several advantages, primarily centered around expediting training time, improving neural network performance (in most instances), and reducing data demands. Traditionally, training a neural network from scratch necessitates a significant volume of data, which may not always be accessible. This is where the value of transfer learning becomes evident. By capitalizing on transfer learning, a robust machine learning model can be developed using a relatively small training dataset, leveraging the model's existing pre-trained state. This is particularly advantageous in fields like natural language processing, where constructing extensive labelled datasets often requires specialized expertise. Furthermore, transfer learning cuts down training time, a crucial benefit given that training complex deep neural networks from the beginning can sometimes span days or even weeks. Nevertheless, it's important to recognize that the effectiveness of transfer learning relies on the

transferability of features learned during the initial task. These features must have relevance to other related tasks as well. Equally essential is the need for input dimensions to match those of the model's original training. If such congruence is lacking, a pre-processing step becomes necessary to adjust the input to the required dimensions.

The ascent of Vision Transformers (ViT) represents a revolutionary approach in the domain of computer vision. This departure has fundamentally transformed the way we approach and dissect visual data. While traditional Convolutional Neural Networks (CNNs) have conventionally dominated visual tasks, ViT presents an innovative alternative. Through the utilization of self-attention mechanisms and Transformer architectures, ViT overcomes the limitations posed by the confined receptive fields in CNNs. This breakthrough empowers ViT to encapsulate extensive global dependencies and far-reaching interactions within images. As a result, this capability drives significant performance enhancements across a range of computer vision tasks, spanning image classification, object detection, and image generation. With their adeptness at modelling intricate, high-dimensional visual data, ViT is reshaping the landscape of computer vision and opening doors to new realms of possibility. To identify the primary antecedent of DR, the authors of [14] suggested automated knowledge models. The model was trained using CNN, DNN, and three times back propagation neural networks. A hybrid model that uses DL and image processing to enhance the outcome for DR prediction. The collection was taken from MESSIDOR and contains 400 images. DL and CNN were used to categorize the diagnosis [15]. Using micro-aneurysms, exudate, and haemorrhage features toidentify DR was the subject of a study in [16].

3. Research Methodology

The task of image classification is to assign an input image to a class or label. The problem involves using a trained model to predict class labels for new, unseen images based on a labelled dataset of images. In the field of image classification, convolutional neural networks (CNNs) are among the most popular architectures. To recognize objects in images, CNNs are particularly effective since they automatically learn features such as edges, textures, and shapes, and the spatial hierarchies of these features. The top layer of a Deep Learning-based model concentrates on extracting more complicated and task-specific data, whereas the bottom layer is in charge of capturing generic information. Some layers of the convolution base are frequently frozen while others are trainable to increase the effectiveness of pre-trained networks. The classification layer is also tailored based on the precise number of classes necessary for the current assignment. In this study, we focused on optimising VGG16 and VGG19 since they outperformed other pre-trained designs in the classification of diabetic retinopathy. The proposed research methodology is shown in figure 1.

The classification layer of the VGG16 network was altered, the number of trainable levels was changed, and new layers were added to increase the network's efficacy. With this strategy, we hoped to help the network pick up more precise elements that were important for our classification task. The modified VGG16 architecture, seen in Figure 3, exemplifies the alteration of the classification layers, the addition of two more levels, and the preservation of the remaining layers by freezing. We have suggested a newcompletely linked Head to improve the VGG19 model. These modifications may be seen in Figure 4's updated structure for VGG19.

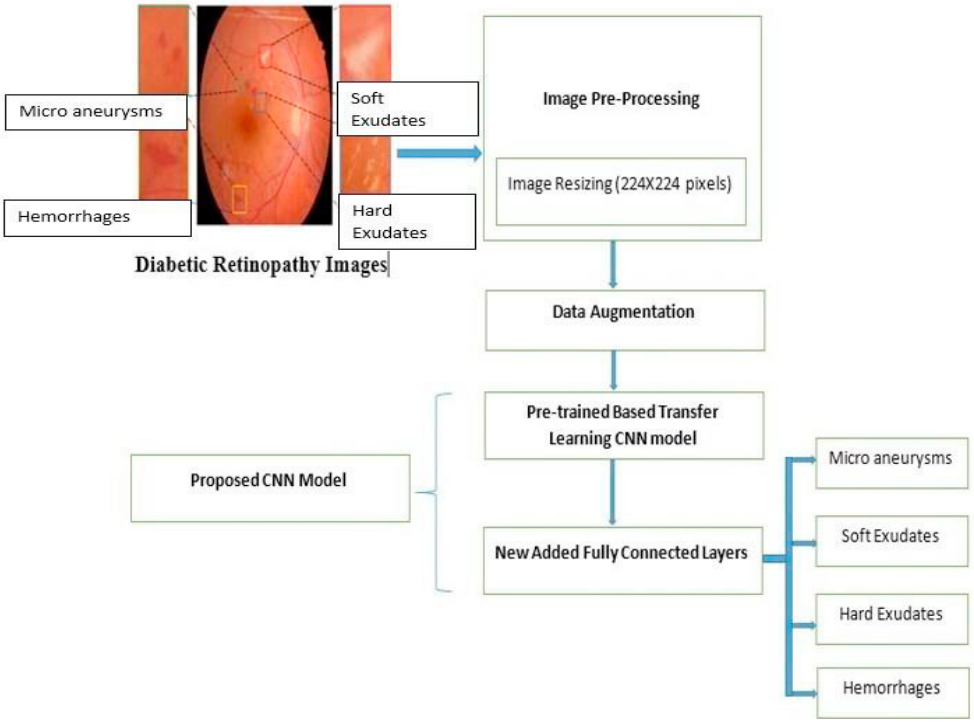


Fig. 1. Author proposed research methodology

3.1. Data Augmentation

Collecting fundus images from multiple datasets is essential, encompassing a diverse array of camera sources. Each image within this collection exhibits distinct perspectives, clarity levels, contrast variations, and image dimensions, all contributing indispensably to heightened precision. The process of data augmentation encompasses a spectrum of enhancements, notably alterations in contrast and image mirroring. Notably, the primary openly accessible Indian Diabetic Retinopathy Image Dataset [17] has found application across multiple research studies. Annotated fundus images showing microaneurysms, hemorrhages, soft exudates, and hard exudates is shown in figure 2. Dataset details are given in table 1. Methods used for data Augmentation with range is shown in table 2.

Table 1. Dataset details

Severity Level of images	Images for Training	Images for Testing	Total used Images
Micro aneurysms (MA) category	54 Images	27 Images	81 Images
Soft Exudates (SE) category	26 Images	14 Images	40 Images
Hard Exudates (EX) category	54 Images	27 Images	81 Images

Haemorrhages (HE) category	53 Images	27 Images	80 Images

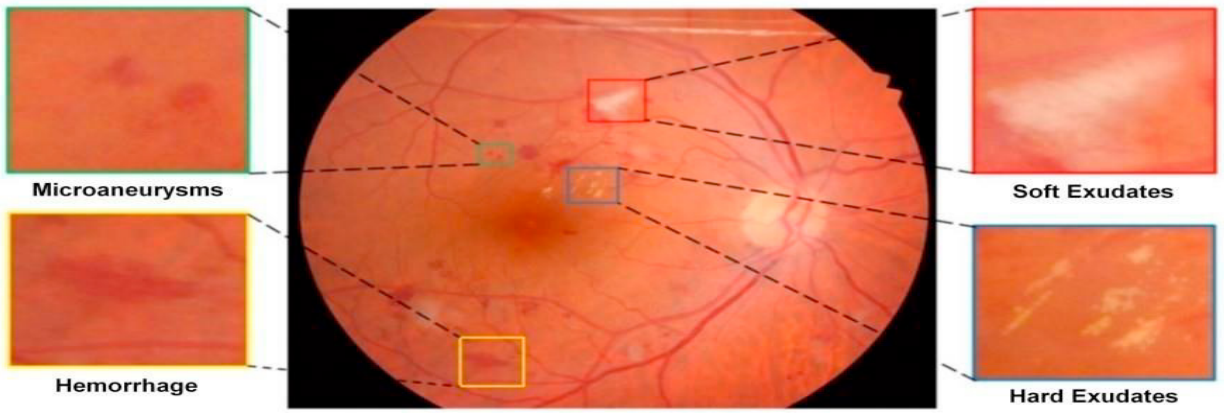


Fig. 2. Annotated fundus images showing microaneurysms, haemorrhages, soft exudates, and hard exudates.

Table 2. Methods used for data Augmentation with range

Augmentation method	Value
Rescale	1/255
Shear range	0.3
Width shift range	0.3
Height shift range	0.3
Zoom range	0.3

3.2. Image Pre-processing

The process of readying images for utilization in model training and inference is referred to as image pre-processing. Various pre-processing techniques like scaling, orienting, and adjusting colours are utilized in this paper. We also reduce the dataset's dimensions, enabling our model to efficiently process it while making optimal use of resources.

3.3. Pre-trained models VGG16 and VGG19

The Visual Geometry Group (VGG) represents a category of pre-trained deep convolutional neural networks (CNNs). The label "deep" pertains to the layer count, with VGG-16 comprising 16 layers and VGG-19 consisting of 19 convolutional layers. Employing the VGG architecture facilitates the development of innovative models for item recognition. Furthermore, VGG remains among the most widely employed and enduring image recognition frameworks to date. Our proposed model uses a default learning rate of 0.001.

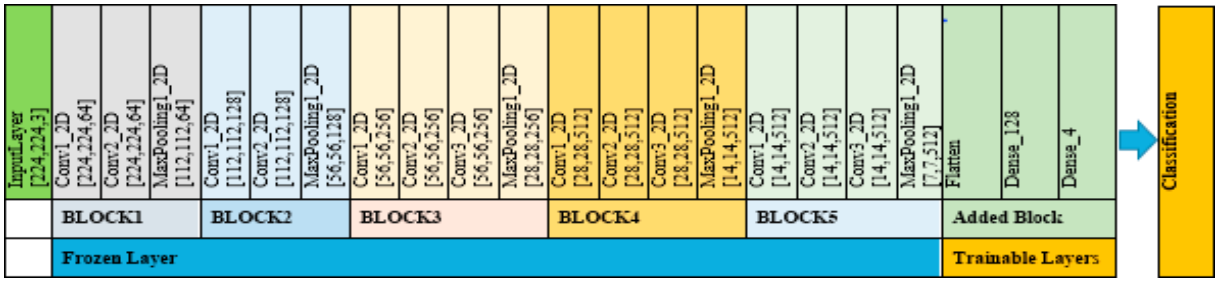


Fig. 3. Author proposed fine-tuned VGG 16 architecture

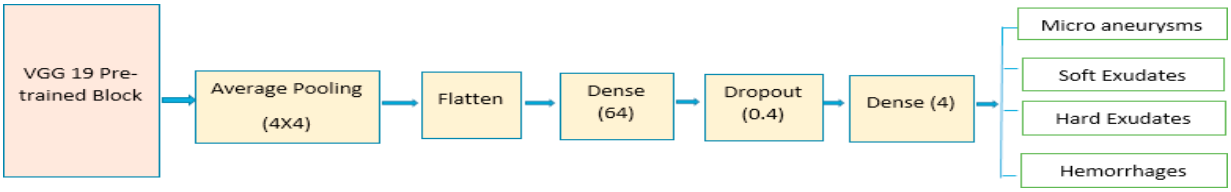


Fig. 4. VGG 19 Proposed New Fully Connected Head

4. Results

4.1. Confusion matrices for the classifiers listed below were generated using VGG-16 feature extraction:

4.1.1. The plotted data illustrates the confusion matrix for Logistic Regression

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	69	3	4	4	80
	Hard Exudates	4	68	8	1	81
	Microaneurysms	0	2	79	0	81
	Soft Exudates	3	1	0	36	40
	∑	76	74	91	41	282

4.1.2. The plotted data illustrates the confusion matrix for Random Forest

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	64	8	3	5	80
	Hard Exudates	5	68	8	0	81
	Microaneurysms	5	1	75	0	81
	Soft Exudates	6	2	0	32	40
	∑	80	79	86	37	282

4.1.3. The plotted data illustrates the confusion matrix for the Neural Network

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	61	9	2	8	80
	Hard Exudates	0	74	6	1	81
	Microaneurysms	2	3	76	0	81

	Soft Exudates	4	1	0	35	40
	∑	67	87	84	44	282

4.1.4. The plotted data illustrates the confusion matrix for SVM

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	67	5	5	3	80
	Hard Exudates	4	66	11	0	81
	Microaneurysms	0	1	80	0	81
	Soft Exudates	5	2	0	33	40
	∑	76	74	96	36	282

4.1.5. The plotted data illustrates the confusion matrix for Gradient Boosting

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	65	6	4	5	80
	Hard Exudates	5	68	7	1	81
	Microaneurysms	1	2	78	0	81
	Soft Exudates	6	1	0	33	40
	∑	77	77	89	39	282

4.1.6. The plotted data illustrates the confusion matrix for AdaBoos

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	55	13	6	6	80
	Hard Exudates	7	61	10	3	81
	Microaneurysms	0	12	69	0	81
	Soft Exudates	13	6	2	19	40
	∑	75	92	87	28	282

Table 3. AUC, CA, F1, Precision and Recall

Model	**AUC**	**CA**	**F1**	**Precision**	**Recall**
SVM	0.975	0.872	0.871	0.876	0.872
Random Forest	0.967	0.848	0.847	0.847	0.848
Neural Network	0.971	0.872	0.871	0.875	0.872
Logistic Regression	0.985	**0.904**	0.903	0.907	0.904

In above Table 3, we can observe that the Logistic Regression model achieved the highest classification accuracy of 90.4 with a 98.5 AUC score, 90.3 F1 score, 90.7 Precision and 90.4 Recall score. So we can select a logistic Regression model to classify diabetic retinopathy images (VGG16 is used as a feature extractor).

4.2. Confusion matrices for the classifiers listed below were generated using VGG-19 feature extraction:

4.2.1. The plotted data illustrates the confusion matrix for Logistic Regression

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	69	3	4	4	80
	Hard Exudates	4	68	8	1	81
	Microaneurysms	0	2	79	0	81
	Soft Exudates	3	1	0	36	40
	∑	76	74	91	41	282

4.2.2. The plotted data illustrates the confusion matrix for Random Forest

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	65	9	3	3	80
	Hard Exudates	11	62	7	1	81
	Microaneurysms	1	5	75	0	81
	Soft Exudates	9	1	0	30	40
	∑	86	77	85	355	282

4.2.3. The plotted data illustrates the confusion matrix for the Neural Network

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	64	8	3	5	80
	Hard Exudates	5	64	9	3	81
	Microaneurysms	0	5	76	0	81
	Soft Exudates	4	1	0	35	40
	∑	73	78	88	43	282

4.2.4. The plotted data illustrates the confusion matrix for SVM

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	62	8	6	4	80
	Hard Exudates	7	58	16	0	81
	Microaneurysms	2	1	78	0	81
	Soft Exudates	0	1	1	38	40
	∑	71	68	101	42	282

4.2.5. The plotted data illustrates the confusion matrix for Gradient Boosting

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	68	3	4	5	80
	Hard Exudates	88	64	88	1	81
	Microaneurysms	1	1	79	0	81
	Soft Exudates	3	1	0	36	40
	∑	80	69	91	42	282

4.2.6. The plotted data illustrates the confusion matrix for AdaBoost

	Predicted					
		Haemorrhages	Hard Exudates	Microaneurysms	Soft Exudates	∑
Actual	Haemorrhages	60	13	1	6	80
	Hard Exudates	10	62	8	1	81
	Microaneurysms	1	4	76	0	81
	Soft Exudates	88	0	0	32	40
	∑	79	79	85	39	282

Table 4. AUC, CA, F1, Precision and Recall

Model	**AUC**	**CA**	**F1**	**Precision**	**Recall**
SVM	0.974	0.837	0.834	0.843	0.837
Random Forest	0.962	0.823	0.822	0.824	0.823
Neural Network	0.975	0.848	0.847	0.848	0.848
Logistic Regression	0.985	**0.894**	0.893	0.895	0.894
Gradient Boosting	0.976	0.876	0.875	0.878	0.876
AdaBoost	0.874	0.816	0.815	0.814	0.816

In above Table 4, we can observe that the Logistic Regression model achieved the highest classification accuracy of 89.4 with a 98.5 AUC score, 89.3 F1 score, 89.5 Precision and 89.4 Recall score. So we can select a logistic Regression model to classify diabetic retinopathy images (VGG19 is used as a feature extractor).

5. Conclusion

In this study, we leveraged our novel fine-tuned models, VGG16 and VGG19, to extract features. Subsequently, we employed a range of machine learning approaches, encompassing both traditional and deep learning methods such as SVM, Random Forest, Neural Networks, Gradient Boosting, and Logistic Regression, for diabetic retinopathy (DR) image classification. Our proposed work achieved its highest accuracy at 90.4%. This exceptional performance was attained when utilizing the proposed VGG16 for feature extraction and coupling it with the Logistic Regression classifier for classification. Notably, our model exhibited the ability to categorize new, unseen images with an impressive accuracy of 90.4%. Looking ahead, we plan to advance our model by integrating contemporary deep-learning techniques, aiming to enhance its

generality and performance. To further elevate the model's effectiveness and ensure a more precise and dependable diagnosis of diabetic retinopathy, future research directions should encompass noise reduction methods, metadata integration, feature selection strategies, and the incorporation of ensemble learning. This comprehensive approach holds promise for augmenting accuracy, reliability, and operational efficiency. Furthermore, the proposed model exhibits strong potential for the detection of diabetic retinopathy disorders in forthcoming endeavours.

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